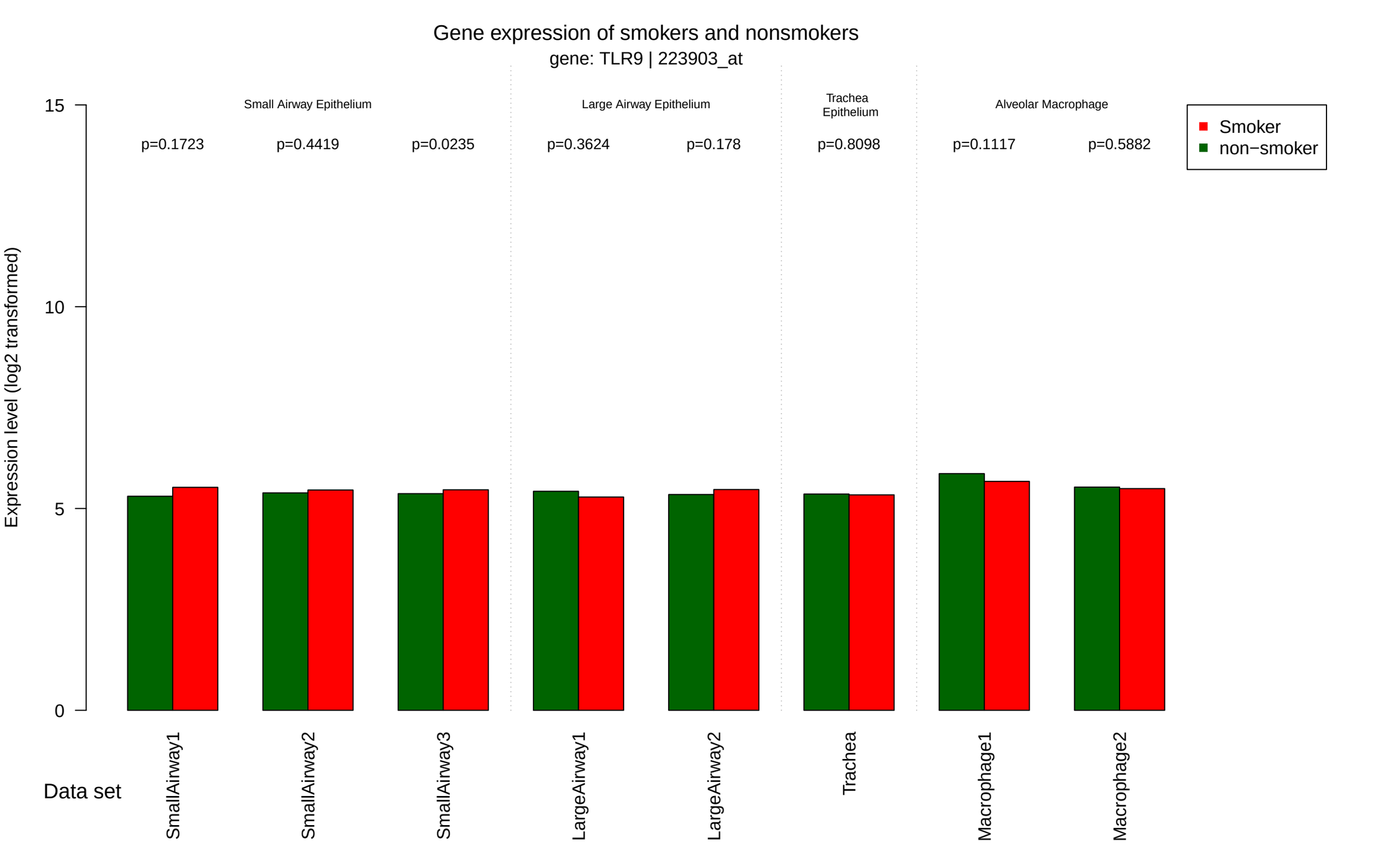
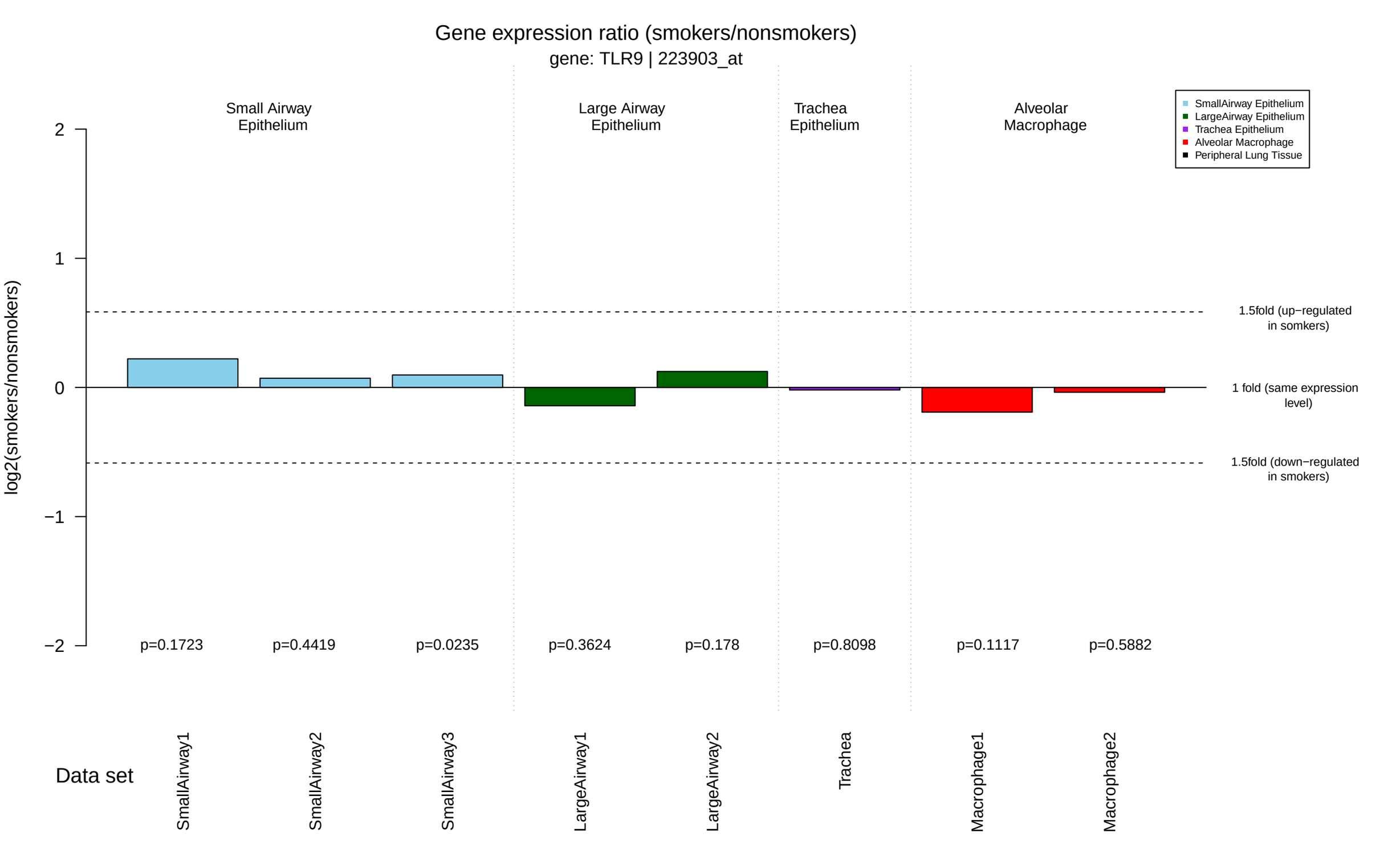
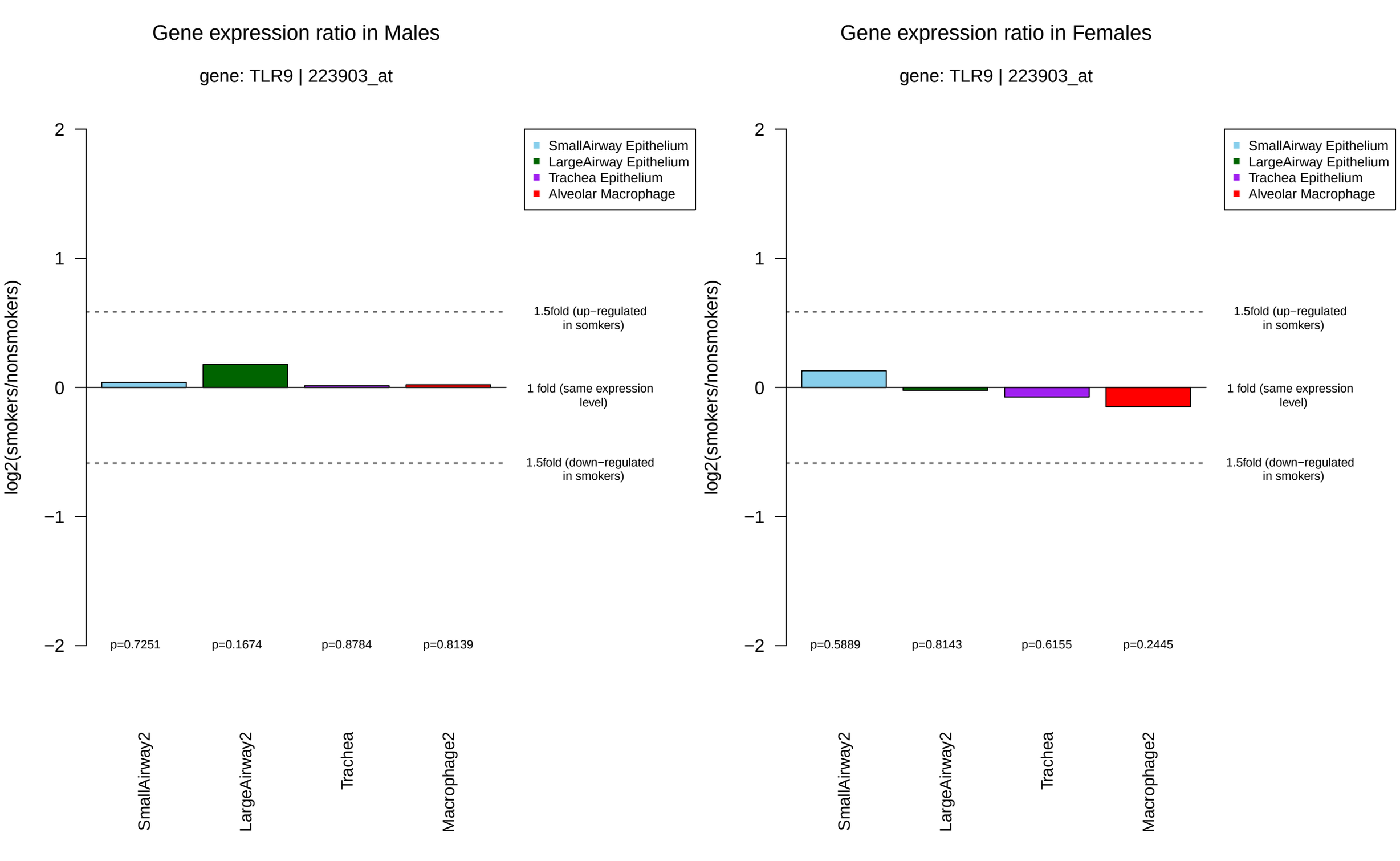
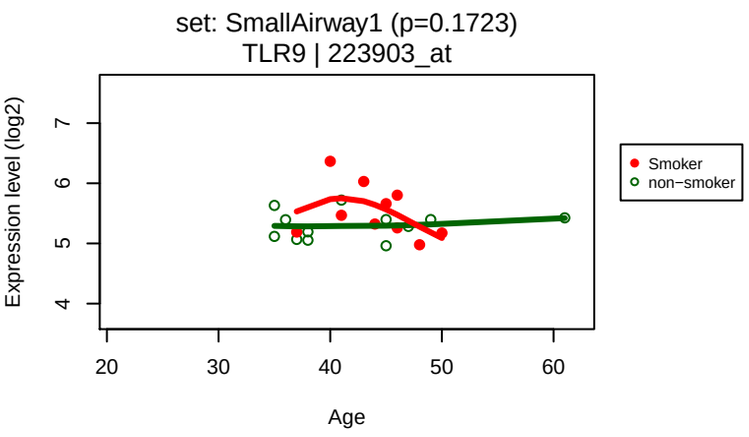
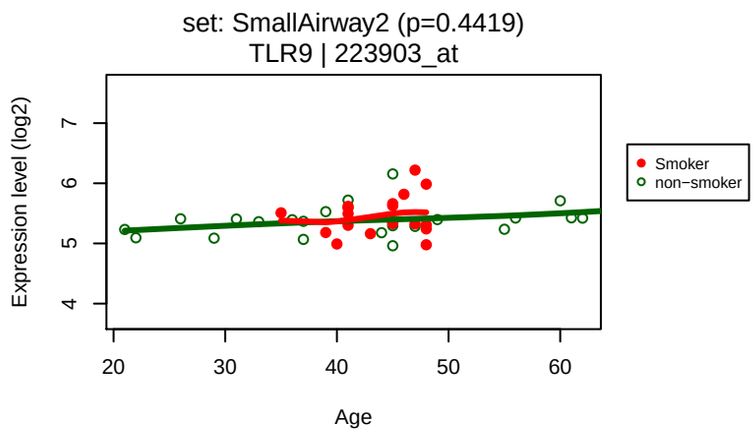
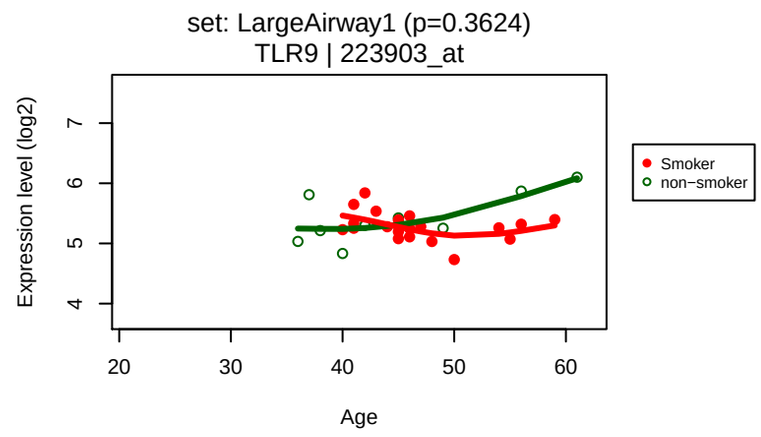

Gene expression of smokers and nonsmokers
gene: TLR9 | 223903_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1723
p=0.4419
p=0.0235
p=0.3624
p=0.178
p=0.8098
p=0.1117
p=0.5882
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

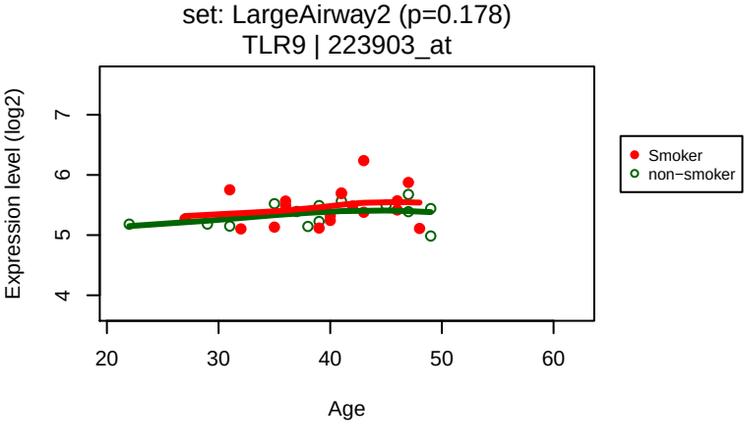

Gene expression ratio (smokers/nonsmokers)
gene: TLR9 | 223903_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1723
p=0.4419
p=0.0235
p=0.3624
p=0.178
p=0.8098
p=0.1117
p=0.5882
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

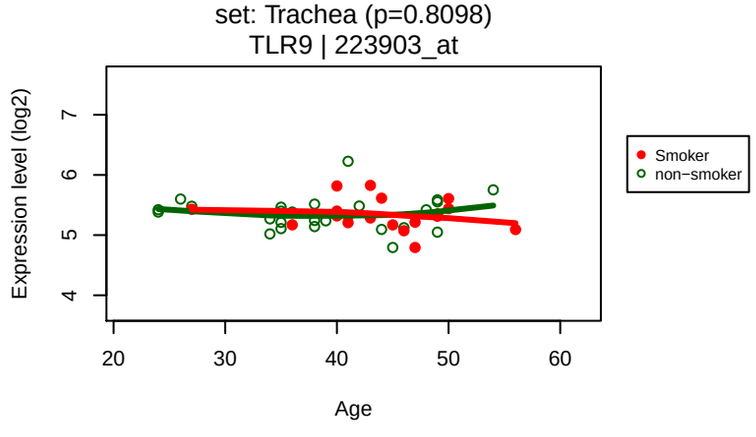

Gene expression ratio in Males
gene: TLR9 | 223903_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7251
p=0.1674
p=0.8784
p=0.8139
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: TLR9 | 223903_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5889
p=0.8143
p=0.6155
p=0.2445
SmallAirway2
LargeAirway2
Trachea
Macrophage2

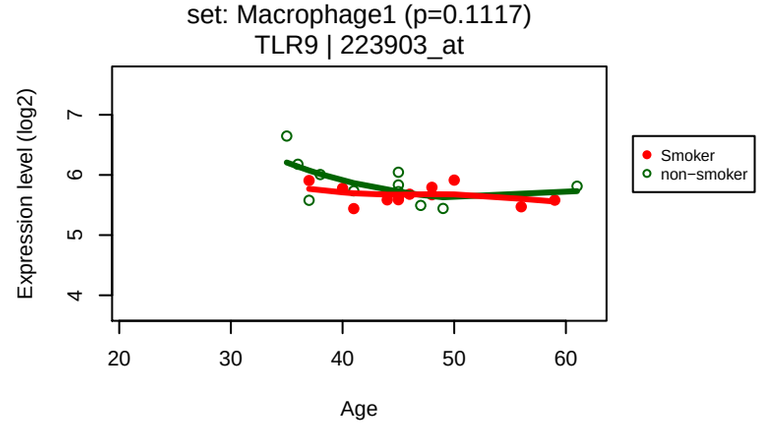

set: SmallAirway1 (p=0.1723)
TLR9 | 223903_at
Expression level (log2)
Age
Smoker
non−smoker

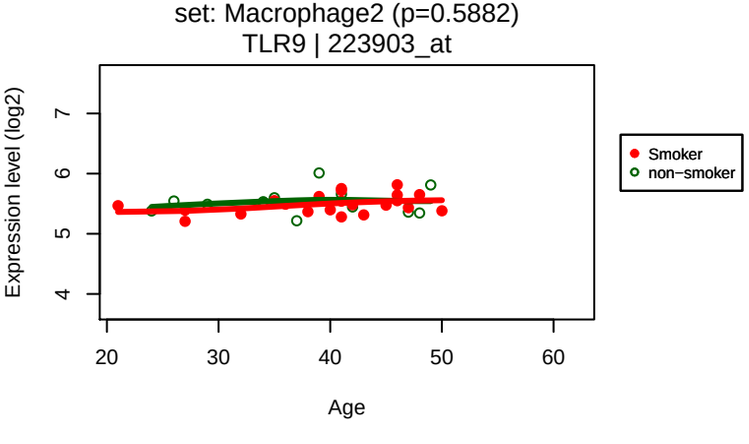

set: SmallAirway2 (p=0.4419)
TLR9 | 223903_at
Expression level (log2)
Age
Smoker
non−smoker

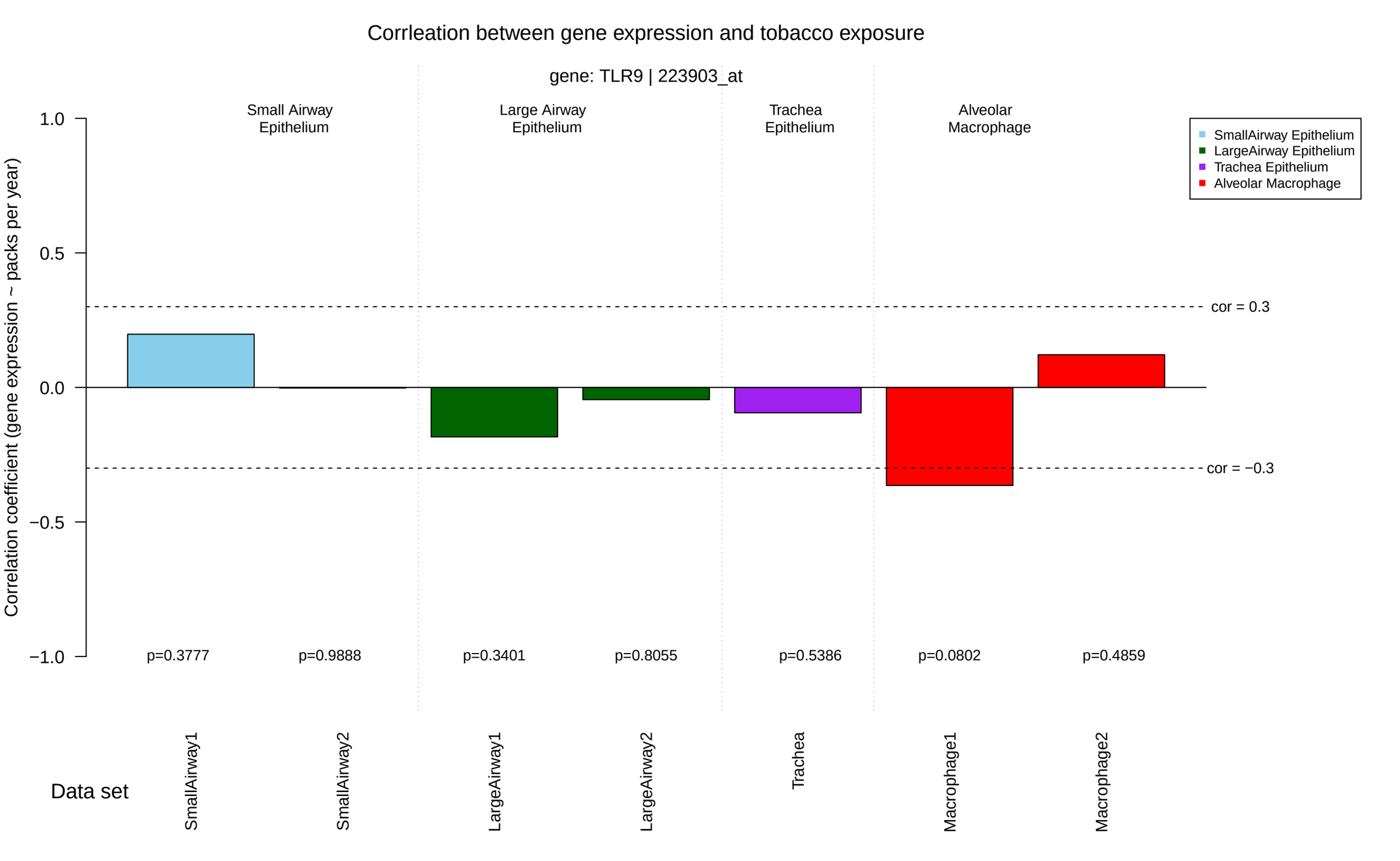

set: LargeAirway1 (p=0.3624)
TLR9 | 223903_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.178)
TLR9 | 223903_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.8098)
TLR9 | 223903_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1117)
TLR9 | 223903_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.5882)
TLR9 | 223903_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: TLR9 | 223903_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.3777
p=0.9888
p=0.3401
p=0.8055
p=0.5386
p=0.0802
p=0.4859
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2